Yiwei Chen

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Education

Michigan State University — Aug. 2024 – Present
Ph.D. in Computer Science, Advisor: Prof. Sijia Liu — *East Lansing, United States*

Xi'an Jiaotong University — Sept. 2021 – Jun. 2024
M.S. in Computer Science and Technology — *Xi'an, China*

Xi'an Jiaotong University — Sept. 2017 – Jun. 2021
B.Eng. in Automation, Young Gifted Class (national early-entrance honors program) — *Xi'an, China*

Research Interests

- **Agents & Post-training:** LLMs, MLLMs, Post-training, Reasoning, Agentic Systems, GUI Agents
- **Trustworthy AI:** AI Safety, Alignment, Trustworthy Algorithms, Interpretability, Machine Unlearning

Internships

Amazon, Applied Scientist Intern — May 2026 – Aug. 2026
Research Project: Memory-Augmented World Models for Mobile GUI Agents — *Sunnyvale, United States*

- Identified the loop-closure problem in GUI world models, where one-step predictors regenerate a revisited page instead of restoring its earlier state, and formulated a trajectory-level prediction task with loop-closure evaluation.
- Built an automated annotation pipeline adding page identity, HTML targets, and revisit labels to three benchmarks.
- Designed a latent memory that stores hidden states of each observed transition, retrieves them when a page is revisited, and injects them into a frozen VLM via zero-initialized cross-attention trained with counterfactual losses.
- Outperformed prior GUI world models on revisit consistency; manuscript in preparation for CVPR 2027.

Cisco, Research Intern — May 2025 – Oct. 2025
Research Project: LLMs for CVE Exploit Generation in Cybersecurity [M.1] — *San Francisco, United States*

- Built a CVE-conditioned exploit generation dataset and an 8-criterion LLM-as-judge evaluation framework.
- Benchmarked 17 open-weight and proprietary LLMs under one setup to characterize their zero-shot capability.
- Fine-tuned Qwen3-8B on the curated data, improving exploit quality by 42.5% and rivaling proprietary models.

Publications

C=Conference, M=Manuscript, *=Equal Contribution

[C.1] **Y. Chen**, Y. Yao, Y. Zhang, B. Shen, G. Liu, S. Liu. Safety Mirage: How Spurious Correlations Undermine VLM Safety Fine-Tuning and Can Be Mitigated by Machine Unlearning. **ICLR'26**.

[C.2] **Y. Chen**, S. Pal, Y. Zhang, Q. Qu, S. Liu. Unlearning Isn't Invisible: Detecting Unlearning Traces in LLMs from Model Outputs. **ICLR'26**.

[C.3] **Y. Chen**, B. Shang, S. Liu. Who Built This Model? Tracing LLM Lineage via Spectral Fingerprints in Weight Space. **COLM'26**.

[C.4] **Y. Chen***, B. Shang*, Y. Zhang, B. Shen, S. Liu. Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning. **COLM'26**.

[M.1] **Y. Chen**, L. Li, K. Cheung, V. Parla, G. Sundaram. Data-Centric Benchmarking of Exploit Generation in LLMs: Understanding the Impact of Fine-Tuning. Manuscript, 2026.

[C.5] **Y. Chen***, Z. Zhang*, W. Zhang, C. Yan, Q. Zheng, Q. Wang, W. Chen. Tile Classification Based Viewport Prediction with Multi-modal Fusion Transformer. **ACM MM'23**.

[C.6] H. Reisizadeh, J. Ruan, **Y. Chen**, S. Pal, S. Liu, M. Hong. Leak@k: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding. **ICML'26**.

[C.7] Y. Zhang, C. Wang, **Y. Chen**, C. Fan, J. Jia, S. Liu. The Fragile Truth of Saliency: Improving LLM Input Attribution via Attention Bias Optimization. **NeurIPS'25 (Spotlight)**.

Research Experience

OPTML Group, Michigan State University Aug. 2024 – Present
Research Assistant, Advisor: Prof. Sijia Liu *East Lansing, United States*

- *Trustworthy Foundation Models:* Investigated the reliability of post-training interventions: characterized spurious correlations in VLM safety fine-tuning and proposed an unlearning-based remedy [C.1]; established that unlearning leaves detectable output-level traces [C.2] and is vulnerable to attention-sink backdoors [C.4]; introduced spectral weight-space fingerprints for model lineage attribution [C.3].
- *Multi-Agent Systems:* Developed a distributed training pipeline for multi-agent LLM systems whose agents reside on separate nodes without checkpoint sharing; cast it as consensus optimization and derived an ADMM algorithm that synchronizes only low-rank adapter states while provably matching centralized training. In preparation for ICLR 2027.
- *GUI Agents:* Investigated long-horizon planning and memory for GUI agents, with a focus on maintaining consistent decision making across extended multi-step interactions with mobile interfaces.

Honors and Awards

Scholarships & Awards

International Conference on Learning Representations, Travel Award	Apr. 2026
Michigan State University, Graduate Travel Award	Oct. 2025
Xi'an Jiaotong University, Outstanding Master Graduate (Top 5%)	Jun. 2024
Xi'an Jiaotong University, Outstanding Bachelor Graduate (Top 5%)	Jun. 2021
Xi'an Jiaotong University, First Prize Scholarship (Top 10%)	2016 – 2019, 2022 – 2023

Competitions

Mathematical Contest in Modeling, Meritorious Winner (Top 8%)	Mar. 2019
China Undergraduate Mathematical Contest in Modeling, First Prize (Shaanxi Division)	Oct. 2018

Services

- **Conference Reviewer:** ICLR, NeurIPS, ACM MM, ICASSP, IJCNN
- **Journal Reviewer:** Journal of Machine Learning Research (JMLR), IEEE Transactions on Signal Processing (TSP)

Skills

- **Programming:** Python, C/C++, Bash, LaTeX
- **LLM Training & Inference:** PyTorch, Hugging Face (Transformers, PEFT, TRL), DeepSpeed, verl, vLLM
- **Models & Agents:** Llama, Qwen, Qwen-VL, LLaVA, LangGraph, MCP
- **Tools:** Git, Docker, Slurm, Claude Code, Cursor, Codex
- **Languages:** Mandarin (Native), English (Professional)